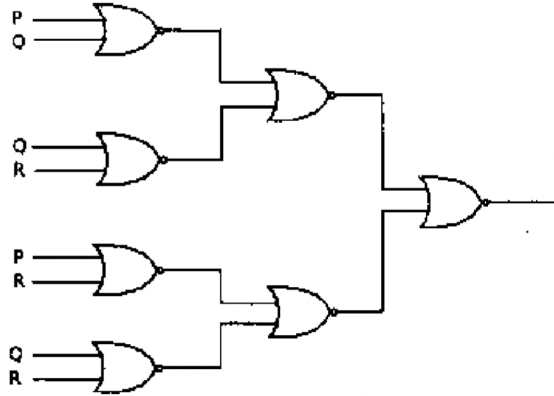


Q.31 What is the boolean expression for the output f of the combinational logic circuit of NOR gates given below?



- (A) $\overline{Q + R}$ (B) $\overline{P + Q}$ (C) $\overline{P + R}$ (D) $\overline{P + Q + R}$

Solution

Since all the gates in the circuit are NOR gates,

$$\text{NOR Gate Output} = \overline{A + B}$$

Let the outputs of the first stage NOR gates be

$$\begin{aligned} X_1 &= \overline{P + Q} \\ X_2 &= \overline{Q + R} \end{aligned}$$

These outputs are applied to another NOR gate.

$$\begin{aligned} Y_1 &= \overline{X_1 + X_2} \\ &= \overline{\overline{P + Q} + \overline{Q + R}} \end{aligned}$$

Using DeMorgan's law,

$$Y_1 = (P + Q)(Q + R)$$

Similarly, for the lower section,

$$\begin{aligned} X_3 &= \overline{P + R} \\ X_4 &= \overline{Q + R} \end{aligned}$$

Their NOR output is

$$\begin{aligned} Y_2 &= \overline{X_3 + X_4} \\ &= \overline{\overline{P + R} + \overline{Q + R}} \end{aligned}$$

Applying DeMorgan's law,

$$Y_2 = (P + R)(Q + R)$$

Finally, the outputs Y_1 and Y_2 are connected to the last NOR gate.

$$\begin{aligned} f &= \overline{Y_1 + Y_2} \\ &= \overline{(P + Q)(Q + R) + (P + R)(Q + R)} \end{aligned}$$

Simplifying,

$$f = \overline{P + R}$$

Correct Answer: Option (C) $\overline{P + R}$